ADI TFLITE MICRO SHARCFX Release Notes

Product Information	
Name	TFLITE-MICRO-SHARCFX
Release Number	3.0.0
Release Date	October 2025
Support	www.analog.com/support

This package contains release 3.0.0 of the ADI TFLITE MICRO SHARCFX product. The ADI TFLITE MICRO SHARCFX is a build of TensorFlow Lite which enables the user to run Deep Neural networks on the SharcFX processors using the optimized TensorFlow Lite Micro Framework. This release is intended for developers who are familiar with the Tensorflow and Tensorflow Lite APIs and framework. This module is intended to serve as a reference with reference applications on how to develop and run your application with the ADI TFLITE MICRO SHARCFX library.

About This Release

ADI TFLITE MICRO SHARCFX 3.0.0 release includes performance enhancements for existing kernels, newly added support for LSTM kernels, headless build and flashing capabilities, multi-boot functionality, and various improvements to the Keyword Spotter, Genre Identification and Urbansound Classification applications.

Your feedback is important to us - please help us make this the best product possible. Keep in mind that we are continuing to work on the ADI TFLITE MICRO SHARCFX and the product may change in the future.

Please note that the models we use in the example applications are for DEMO purpose only. If a product needs to be created using the same models, commercial license needs to be taken, and the necessary due diligence needs to be handled by the party with the rightful owners.

Release 3.0.0 Highlights

- Performance improvements for Fully connected kernels, Tanh and Sigmoid [Logistic] kernels.
- New optimized kernels added:
    - LSTM
    - Elementwise add
    - Elementwise multiply.
- Support for headless build.
- Support for multi-application boot from flash.

- Improvements in applications:
    - Keyword Spotter: Input playback added
    - Genre Identification: Realtime support added
    - Urbansound Classification: File IO and Realtime support added

Release 2.2.0 Highlights

- This release separates the previous optimized version of TFLM library for SharcFX processors into 2 separate libraries – TFLM and adi_sharcfx_nn [contains the optimized kernels].
- Performance improvements for Depthconv2D and Conv2D optimized kernels.
- Code refactoring making it more user-friendly.
- Prebuilt flash-able .ldr files for denoiser_realtime and kws_realtime applications.

Release 2.1.0 Highlights

- This release contains the optimized version of TFLM library for SharcFX processors.
- The following reference applications are added in this release.
    - Keyword spotter
        - Realtime input

Release 2.0.0 Highlights

- This release contains the optimized version of TFLM library for SharcFX processors.
- The following reference applications are supported:
    - Keyword spotter
        - File io input
    - Denoiser
        - File io input
        - Realtime input via ADC
    - Genre identification
        - File io input
        - Realtime input via ADC
    - Urbansound Classification
        - File io input
        - Realtime input via ADC
    - People detection
    - Microspeech
    - Hello world

Release 1.0.2 Highlights

- Integration and Unit Test projects for Microspeech and People Detect applications.
- Both Microspeech and People Detect are matching to ref implementation now.
- Macro allows to switch between 40 bit accumulator and 80 bit accumulator in primitives for higher accuracy.

Release 1.0.1 Alpha Highlights

- Further optimizations for Convolutional and Depthconv kernels for People Detection application.
- MicroSpeech example also includes optimized fully connected and depthconv kernel support.
- A test project for People Detection is included in this release along with various test cases.

Release 1.0.0 Alpha Highlights

- This package contains the TensorFlow Lite Micro sources and library compiled with the Eagle HP compiler.
- 3 example applications – People Detection, Micro Speech and Hello World are also included.
- People Detection application includes optimized kernels for the SharcFX processor

Installation Information

This package is released as zip package and can be unzipped and installed in any folder. Elsewhere in this document this default path is referred to as ***<SWM_ROOT>***. This package is unzipped into ***<SWM_ROOT>\ADI_TFLITE_MICRO_SHARCFX_PROD-Rel3.0.0***.

Known Issues and Limitations

Known Issues

None

Technical Support:

Contact your local FAE for support on this product.